

A. Maximum Neighborhood

time limit per test: 2 seconds

memory limit per test: 512 megabytes

Consider an $n \times n$ grid filled with numbers as follows:

- the first row contains integers from 1 to n from left to right;
- the second row contains integers from $(n + 1)$ to $2n$ from left to right;
- this pattern continues until the n -th row, which contains integers from $(n^2 - n + 1)$ to n^2 from left to right.

Let's define the cost of a cell as its value plus the sum of its neighboring cells' values. Two cells are considered neighboring if they share a side.

Your task is to calculate the maximum cost among all cells in the grid.

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

The grid for $n = 4$ and the optimal answer for it. The yellow cell has the maximum possible cost; the green cells are its neighbors. The cost of the cell is $15 + 11 + 14 + 16 = 56$.

Input

The first line contains a single integer t ($1 \leq t \leq 100$) — the number of test cases.

The only line of each test case contains a single integer n ($1 \leq n \leq 100$).

Output

For each test case, print a single integer — the maximum cost among all cells in the grid.

Example

input	Copy
5 1 2 3 4 5	
output	Copy
1 9 29 56 95	

Note

In the first example, there is only 1 cell with the cost 1.

In the second example, the cell with value 4 has the maximum cost: $4 + 2 + 3 = 9$.

In the third example, the cell with value 8 has the maximum cost: $8 + 5 + 7 + 9 = 29$.

Educational Codeforces Round 185 (Rated for Div. 2)

Contest is running

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Contestant


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 Language: Python 3.13.2 
 Almost always, if you send a solution on PyPy, it works much faster

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In the fourth example, the cell with value 15 has the maximum cost: $15 + 11 + 14 + 16 = 56$.

In the fifth example, the cell with value 19 has the maximum cost:
 $19 + 14 + 18 + 20 + 24 = 95$.

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